Math 526, F06 **Name:**

Quiz 6 **SSN: xxx-xx-**

<u>Instructions:</u> **There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.**

Problem 1. (14 points) Let $\mathbf{x} = \begin{bmatrix} 1 \\ -2 \\ 2 \\ 7 \end{bmatrix}$, $\mathbf{A} = \begin{bmatrix} -1 & 2 & 3 & 1 \\ 2 & 3 & -4 & 1 \\ 2 & -5 & -2 & 1 \\ -1 & -1 & -1 & 2 \end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix}$.

a) Compute $||\mathbf{x}||_1$, $||\mathbf{x}||_2$ and $||\mathbf{x}||_\infty$.

Solution:

$$||\mathbf{x}||_1 = 1 + 2 + 2 + 7 = 12.$$

$$||\mathbf{x}||_2 = \sqrt{1^2 + (-2)^2 + 2^2 + 7^2} = \sqrt{58}.$$

$$||\mathbf{x}||_\infty = \max\{1, 2, 2, 7\} = 7.$$

b) Compute $||\mathbf{A}||_1$,$||\mathbf{A}||_\infty$ and $||\mathbf{B}||_2$.

Solution:

$$||\mathbf{A}||_1 = \max\{1+2+2+1, 2+3+5+1, 3+4+2+1, 1+1+1+2\} = 11.$$

$$||\mathbf{A}||_\infty = \max\{1+2+3+1, 2+3+4+1, 2+5+2+1, 1+1+1+2\} = 10.$$

$$||\mathbf{B}||_2 = \max\{1, 3, 5\} = 5.$$

Problem 2. (6 points) Describe in details the three basic properties of matrix norm: *positivity*, *homogeneity* and *triangle inequality*.

Solution:

Positivity: $||\mathbf{A}|| \geq 0$ and $||\mathbf{A}|| = 0$ if and only if $\mathbf{A} = \mathbf{0}$.

Homogeneity: $||\alpha\mathbf{A}|| = |\alpha|||\mathbf{A}||$ for any scalar α.

Triangle inequality: $||\mathbf{A} + \mathbf{B}|| \leq ||\mathbf{A}|| + ||\mathbf{B}||$.